Demo: From Observed Treasury Yields to Hull–White Future Curves

A hands-on companion to Lecture 7

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

1 Goal

This demo follows one complete fixed-income workflow:

1. start with a **dated U.S. Treasury par-yield snapshot**;
2. bootstrap today's discount factors and zero-coupon yield curve;
3. calibrate Hull–White so that it exactly respects that curve;
4. fit the model's mean reversion and volatility to a supplied synthetic option quote snapshot that demonstrates the calibration mechanics;
5. generate yield curves at a future date in different interest-rate states; and
6. use those future curves to price remaining bond cash flows.

The initial curve uses the U.S. Treasury's published **January 2, 2025** par-yield snapshot. To keep every calculation auditable, the market nodes are embedded below rather than downloaded during rendering. Treasury par yields are bond-equivalent quoted rates; bootstrapped zero rates are continuously compounded. The option premiums used later are explicitly labeled synthetic because a stable, public option-volatility surface is not available with the course files.

Source: U.S. Treasury, Daily Treasury Par Yield Curve Rates.

> The central idea
>
> Today's curve determines today's prices. Hull–White adds a disciplined way to describe *future* curves. Calibration makes those two statements consistent.

2 Setup

2.1 Read the workflow in two layers

Valuation today: dated Treasury par yields → constructed annual-pay par coupon bonds → bootstrapped discount factors → present value of fixed cash flows. These constructed instruments are a curve-building convention, not individual Treasury securities or their actual coupons.

Uncertainty tomorrow: supplied synthetic option premiums → fitted mean reversion a and volatility σ → curve-consistent drift $\theta(t)$ → future short-rate state r_t → future discount curve → remaining bond value. The premiums illustrate calibration; they are not observed option quotes. The second layer must preserve the first. A complicated model is not allowed to change today's price of fixed cash flows once the discount curve is fixed.

Open **Show calculation** to inspect each step. The final state explorer changes the future state while holding the calibrated parameters and initial curve fixed.

We only need NumPy, pandas, and Matplotlib. The functions below use explicit inputs and return ordinary arrays or data frames so that each step is easy to inspect.

3 Record the observed Treasury curve

The published Treasury series contains par yields at standard maturity nodes. We retain the 1-, 2-, 3-, 5-, 7-, and 10-year observations and linearly interpolate missing integer maturities. The interpolation is an **estimated curve-construction choice**, not an additional market quote. For transparent annual bootstrapping we treat each interpolated par yield as the coupon on an annual-pay par bond. A production Treasury curve would instead use security-level prices, semiannual coupons, actual day counts, accrued interest, and bills at the short end.

Observation date	Maturity (years)	Observed Treasury par yield (%)	Annual par coupon rate (%)	Const
2025-01-02	1	4.17	4.17	100
2025-01-02	2	4.25	4.25	100
2025-01-02	3	4.29	4.29	100
2025-01-02	4		4.335	100
2025-01-02	5	4.38	4.38	100
2025-01-02	6		4.425	100
2025-01-02	7	4.47	4.47	100
2025-01-02	8		4.503333	100
2025-01-02	9		4.536667	100
2025-01-02	10	4.57	4.57	100

Observed nodes are facts from the dated snapshot. Interpolated nodes and the annual-pay par-bond representation are assumptions. That separation matters: a curve can fit its inputs perfectly while still depending on construction choices.

4 Bootstrap today's discount curve

For the one-year bond, the only cash flow is coupon plus principal, so its price immediately gives $P(0,1)$. For each later bond, discount factors for earlier coupons are already known. The final discount factor is the only unknown:

$$P(0,n) = \frac{100 - \sum_{i=1}^{n-1} C_n P(0,i)}{100 + C_n}, \qquad C_n = 100 c_n.$$

Maturity (years)	Discount factor P(0,T)	Continuous zero rate (%)
1	0.959969	4.085399
2	0.920097	4.163799
3	0.881528	4.203298
4	0.84371	4.248659
5	0.806752	4.294781
6	0.770665	4.3417
7	0.735457	4.389466
8	0.701877	4.424966
9	0.669306	4.461266
10	0.637732	4.498374

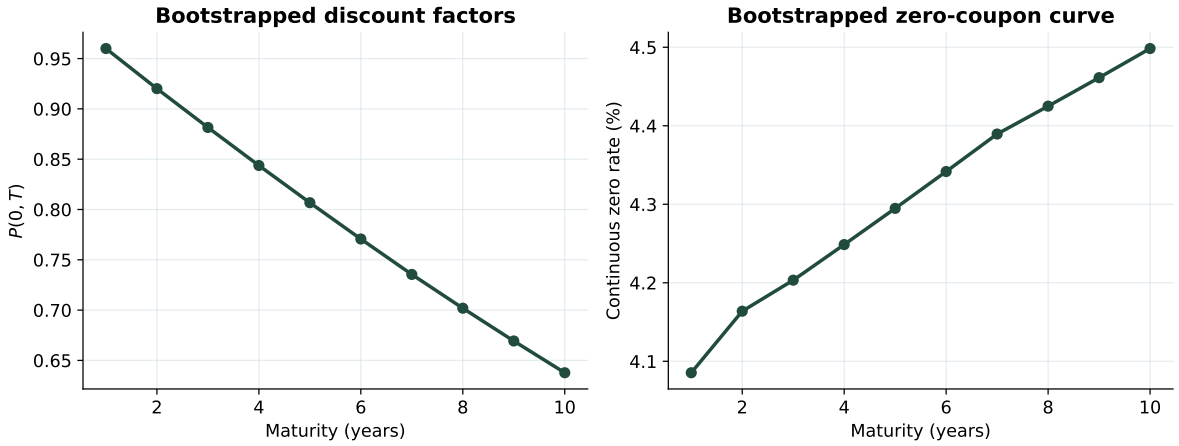


Figure 1: Today's curve recovered from constructed par coupon bonds

4.1 Check the bootstrap by repricing every input bond

A useful calibration habit is to return to the original observable quantities.

Maturity (years)	Annual par coupon rate (%)	Constructed price per 100	Input status	Repri
1	4.17	100	Observed Treasury node	100
2	4.25	100	Observed Treasury node	100
3	4.29	100	Observed Treasury node	100
4	4.335	100	Interpolated node	100
5	4.38	100	Observed Treasury node	100
6	4.425	100	Interpolated node	100
7	4.47	100	Observed Treasury node	100
8	4.503333	100	Interpolated node	100
9	4.536667	100	Interpolated node	100
10	4.57	100	Observed Treasury node	100

```
Largest absolute repricing error: 0.0000000000 per 100 face
```

5 Make the initial curve usable at any maturity

i Bootstrap two maturities by hand

Under the annual-pay par-bond convention, the 1-year discount factor is $P(0,1) = 100/104.17$. The 2-year coupon is 4.25, so $P(0,2) = [100 - 4.25P(0,1)]/104.25$. We remove the present value of the earlier coupon before solving for the final payment's discount factor. Using $1/(1+4.25\%)^2$ instead would confuse the par yield with a zero yield.

There are two different interpolations. Before bootstrapping, we interpolate **par yields** to construct annual-pay par instruments at missing integer maturities. After bootstrapping, we have discount factors $P(0,T_i)$ and continuously compounded zero rates $y(0,T_i) = -\log P(0,T_i)/T_i$. Here we linearly interpolate those **zero rates**, not par yields or discount factors.

For any nonnegative maturity T, this supplies

$$P(0,T) = e^{-y(0,T)T}, \qquad f(0,t) = -\frac{\partial \log P(0,t)}{\partial t}.$$

The first zero rate is held constant between time 0 and year 1; beyond year 10, the last zero rate is held constant. These interpolation and extrapolation choices add no observed market information. Rates are stored as decimals: 0.0417 means 4.17%; displayed percent columns multiply by 100.

The instantaneous forward is approximated by a finite difference of $\log P(0,t)$ with a 10^{-4}-year step, centered except near zero. Piecewise-linear zero rates can produce forward jumps at

maturity knots; differentiating those forwards for $\theta(t)$ creates sharp numerical drift changes. A smooth curve would be needed for stable continuous-time simulation.

6 Fit the Hull–White dynamics to option prices

The lecture's calibration logic separates two jobs:

- the function $\theta(t)$ makes the model fit today's *entire curve*; and
- a and σ control future mean reversion and volatility.

In practice, a and σ are fitted to liquid caps, floors, or swaptions. To keep the code transparent, our artificial option market consists of European calls on zero-coupon bonds. These option premiums could be observed in a suitable market dataset; the premiums below are synthetic. Hull–White has a closed-form pricing formula for them.

6.1 Hull–White building blocks

For the one-factor model

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t^{\mathbb{Q}},$$

t is the valuation time and $T > t$ is a payment maturity, both measured in years. r_t is the instantaneous short rate; a is mean reversion (per year), σ is short-rate volatility (per square root of a year), and $\theta(t)$ is the time-dependent drift intercept. $W_t^{\mathbb{Q}}$ is Brownian motion under the risk-neutral pricing measure. Lowercase a is a scalar parameter; uppercase $A(t,T)$ below is a bond-pricing coefficient, not that parameter.

The maturity loading $B(t,T)$, measured in years, is

$$B(t,T) = \frac{1 - e^{-a(T-t)}}{a}.$$

The volatility of a zero-coupon bond option expiring at S on a bond maturing at T is

$$\sigma_P = \sigma B(S,T)\sqrt{\frac{1 - e^{-2aS}}{2a}}.$$

6.2 Create a synthetic option quote snapshot

We create at-the-money-forward calls with expiries from six months to five years. The hidden values $a = 0.18$ per year and $\sigma = 1.20\%$ per square root of a year are used only to generate the synthetic premiums. A small alternating bid/ask-like perturbation prevents a mechanically perfect parameter recovery.

Option expiry (years)	Bond maturity (years)	Strike per 100	Synthetic option premium per 100
0.5	5.5	80.48686	0.84443
1	6	80.28013	1.11236
1.5	6.5	80.10003	1.28184
2	7	79.93258	1.3817
3	8	79.62052	1.50171
4	9	79.32891	1.53621
5	10	79.04931	1.53548

6.3 Calibrate a and σ with a simple grid search

The objective is the root-mean-square pricing error across the option quotes. A grid search is slower than a numerical optimizer, but students can see exactly what is tried and it needs no extra package.

```
Calibrated mean reversion a: 0.1790 per year
Calibrated volatility sigma: 1.1919% per sqrt(year)
Option-price RMSE: 0.005442 per 100 face
```

Option expiry (years)	Bond maturity (years)	Strike per 100	Synthetic option premium per 100	Model p
0.5	5.5	80.486861	0.844433	0.838257
1	6	80.280132	1.11236	1.110013
1.5	6.5	80.100034	1.281842	1.274304
2	7	79.932583	1.381698	1.380736
3	8	79.620518	1.501708	1.495199
4	9	79.328909	1.536209	1.533078
5	10	79.049312	1.535476	1.528147

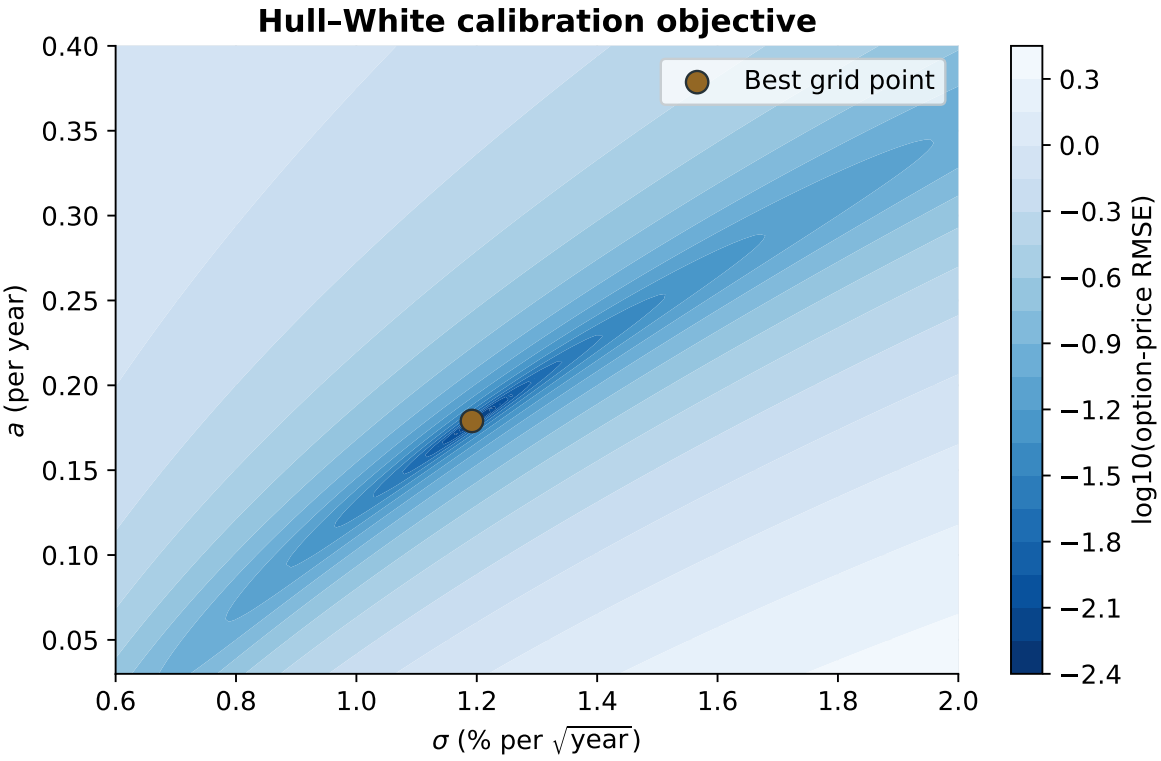


Figure 2: Option-pricing error across candidate Hull–White parameters

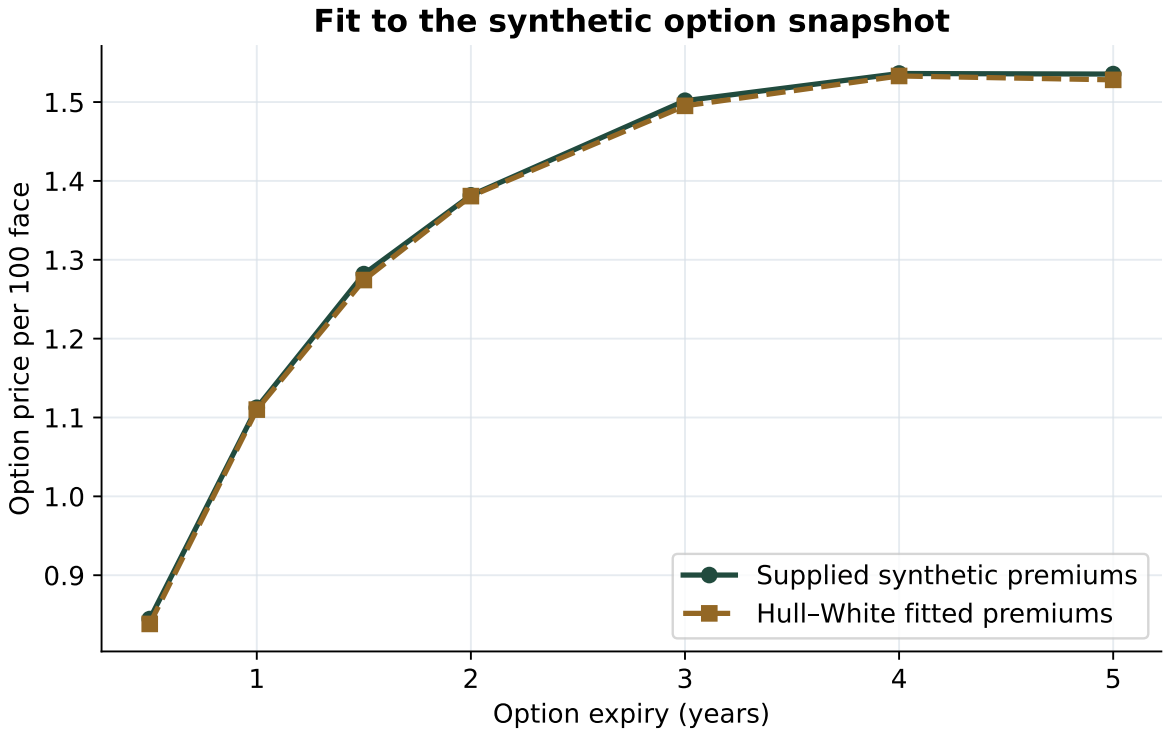


Figure 3: Observed artificial option premiums and calibrated Hull–White prices

7 Fit the time-dependent drift to today's curve

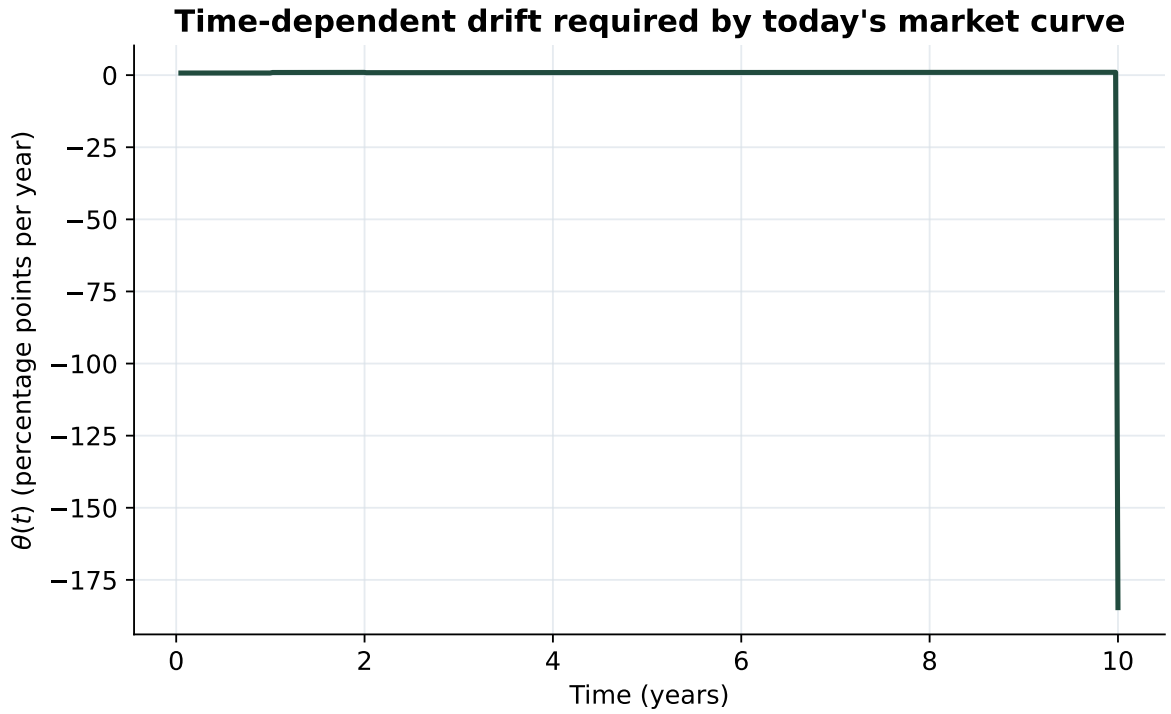
Once a and σ are known, the drift function that fits the observed initial curve is

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + af(0,t) + \frac{\sigma^2}{2a}\left(1 - e^{-2at}\right).$$

The initial curve and the option calibration do different jobs. Fixing a and σ specifies how rates revert and fluctuate; it does not give enough freedom to fit every initial maturity. The time-dependent function $\theta(t)$ supplies that freedom. It is determined from the initial forward curve once a and σ are fixed, rather than fitted as one extra constant.

The forward-slope term follows the changing shape of today's curve. The $af(0,t)$ term offsets mean reversion, and the volatility term adjusts for stochastic discounting. Together they make risk-neutral dynamics consistent with today's discount factors. In this convention the drift is $\theta(t) - ar_t$; $\theta(t)$ itself is not a long-run rate level.

The code approximates the forward slope using a 10^{-3}-year finite difference. The sharp knot behavior described above limits the smoothness of this numerical drift. The future bond-pricing formula below uses the initial discount curve directly, avoiding numerical integration of this plotted drift.



The sharp changes are a consequence of using simple piecewise-linear interpolation. That is instructive: curve construction and smoothing matter in real calibration.

7.1 Verify that Hull–White reproduces today's discount curve

A zero-coupon bond pays one unit at maturity. Its conditional price averages the discounted payoff over all future paths from the current information $\mathcal{F}_t$:

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s\, ds\right) \mid \mathcal{F}_t\right].$$

Hull–White evaluates this expectation in closed form:

$$P(t,T) = A(t,T)e^{-B(t,T)r_t},$$

where

$$A(t,T) = \frac{P(0,T)}{P(0,t)} \exp\left[B(t,T)f(0,t) - \frac{\sigma^2}{4a}(1-e^{-2at})B(t,T)^2\right].$$

$A(t,T)$ incorporates the initial discount curve, model parameters, and dates. $B(t,T)$ loads the simulated short-rate state into the price. A single r_t is sufficient only after the initial curve, parameters, and dates are specified. This formula integrates over subsequent rate paths; it does not assume rates stay at r_t and does not require nested simulation.

At $t = 0$, the volatility adjustment vanishes and $P(0,0) = 1$, so the model expression becomes

$$P(0,T)\exp[B(0,T)f(0,0)]\exp[-B(0,T)r_0] = P(0,T)$$

when $r_0 = f(0,0)$. Thus every maturity of the constructed initial curve is reproduced, not just a particular bond price. The table compares the formula with bootstrapped integer-maturity inputs; the same cancellation holds between nodes under the chosen interpolation rule.

Maturity (years)	Initial P(0,T)	Hull–White P(0,T)	Difference
1	0.959969	0.959969	0
2	0.920097	0.920097	0
3	0.881528	0.881528	-0
4	0.84371	0.84371	0
5	0.806752	0.806752	0

Maturity (years)	Initial P(0,T)	Hull–White P(0,T)	Difference
6	0.770665	0.770665	0
7	0.735457	0.735457	0
8	0.701877	0.701877	0
9	0.669306	0.669306	0
10	0.637732	0.637732	0

8 Generate future yield curves

Move to year 2. In a one-factor Hull–White model, the short rate r_2 is the single state variable. Hold the initial curve, a, and σ fixed. For a simulated or selected r_2, calculate $B(2,T)$ and $A(2,T)$ at each future payment date, then obtain $P(2,T) = A(2,T)e^{-B(2,T)r_2}$. Convert each conditional bond price to its continuously compounded zero yield:

$$y(2,T) = -\frac{\log P(2,T)}{T-2}.$$

We show three possible states around the model's central short-rate level. They are scenarios, not point forecasts.

The center below is the initial instantaneous forward rate at year 2, chosen as a reference. It is not claimed to be the physical forecast or even the exact risk-neutral expected short rate; the latter also contains a volatility adjustment.

Check	Rate state	Short rate (%)	Maturity (years)	Continuous zero rate (%)
0	Low-rate state	2.782	2.25	2.788
1	Low-rate state	2.782	2.3481	2.806
2	Low-rate state	2.782	2.4462	2.823
80	Central state	4.282	2.25	4.255
81	Central state	4.282	2.3481	4.26
82	Central state	4.282	2.4462	4.265
160	High-rate state	5.782	2.25	5.722
161	High-rate state	5.782	2.3481	5.714
162	High-rate state	5.782	2.4462	5.706

Notice that one factor moves the whole curve in a structured way. Mean reversion causes the effect of today's short-rate state to weaken at longer maturities because the zero-yield sensitivity is $B(t,T)/(T-t)$: $B(t,T)$ approaches $1/a$ while remaining maturity grows.

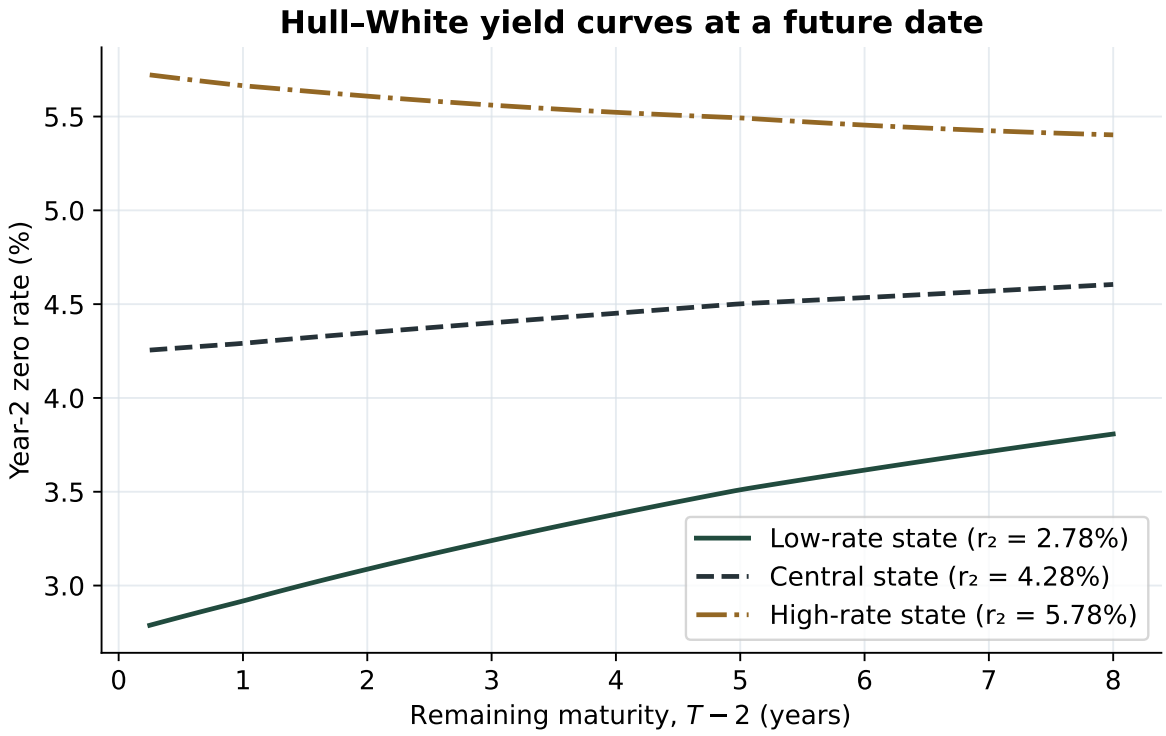


Figure 4: At year 2, one short-rate state determines a complete remaining yield curve

9 Use a future curve to price a bond

Consider the original ten-year 4.10% coupon bond at year 2, immediately after its year-2 coupon has been paid. Eight coupons and the principal remain. Its value in each state is simply the discounted value of those remaining cash flows using $P(2,T)$.

Rate state	Short rate (%)	Year-2 bond value per 100
Low-rate state	2.7817	101.8046
Central state	4.2817	96.106
High-rate state	5.7817	90.7425

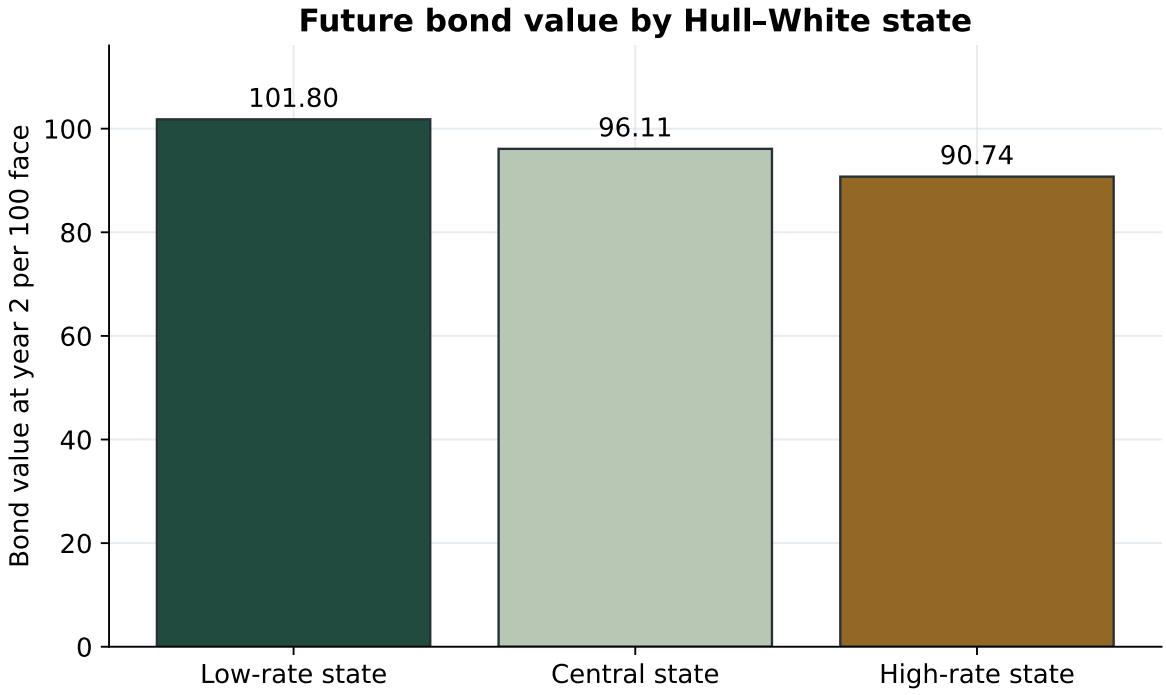


Figure 5: The same promised cash flows have different values in different year-2 rate states

This future valuation is where a short-rate model becomes useful. Today we do not observe $P(2,T)$; Hull–White generates it consistently for each possible r_2.

10 Plain-bond consistency at time 0

Finally, verify the lecture's important claim: once Hull–White is calibrated to the observed curve, it does **not** create a different time-0 price for a vanilla bond. Both methods discount

the same fixed cash flows with the same $P(0,T)$.

```
Direct observed-curve price: 96.2742663687
Calibrated Hull-White price:  96.2742663687
Difference:                    0.000000000000
```

The equality is structural, not a coincidence. At time 0, the calibrated model uses the market discount factors by construction. For a vanilla bond, the model adds no new pricing information. Its value is the ability to generate future curves and to value cash flows that depend on those future states.

11 Checks and interpretation

The workflow has three separate fits, each with a different purpose:

Calibration object	Observable input	What is fitted	Why it matters
Today's discount curve	Coupon-bond prices	$P(0,T)$	Prices fixed cash flows today
Initial Hull–White term structure	Bootstrapped $P(0,T)$	$\theta(t)$	Makes the model no-arbitrage and curve-consistent
Future dynamics	Interest-rate option premiums	a, σ	Controls mean reversion and uncertainty

Check	passed
Bootstrap reprices input bonds	True
Discount factors are positive	True
Discount factors decrease with maturity	True
Hull–White matches today's discount curve	True
Two vanilla-bond pricing methods agree	True

12 Takeaways

> **Separate calibration from a scenario**
>
> First compare observed and interpolated nodes. Then inspect the option-fit error: an exact fit to today's curve does not guarantee an exact fit to every option. Finally change only the future short-rate state. Today's curve stays unchanged, but the future bond value changes. Explain why the year-2 value excludes coupons already paid and is not a two-year holding-period total return.

Market handoff: the embedded Treasury observations are a reproducible dated snapshot, not today's curve. Replace the entire dated node table together, retain the observed/interpolated labels, and rerun every check. Keep the supplied synthetic option prices if a permitted option dataset is unavailable; do not label recovered parameters as empirically calibrated market dynamics.

- Market bond prices can be converted into discount factors by bootstrapping.
- Hull–White does not fit a whole curve using only two constants. The function $\theta(t)$ fits today's term structure; a and σ describe future dynamics.
- Option prices provide information about those dynamics that vanilla bond prices do not.
- At a future date, each one-factor Hull–White short-rate state produces a complete yield curve and a value for the remaining cash flows.
- A vanilla bond has the same time-0 value under calibrated Hull–White and direct curve discounting. The model earns its keep when future rates affect valuation or cash flows.

> **Simplifications**
>
> Real desks use multiple instrument types, day-count conventions, accrued interest, bid/ask spreads, smooth curve interpolation, and robust numerical optimizers. The demo deliberately removes those layers so the economic logic stays visible.

13 Interactive explorer

Change the year-2 short rate relative to the reference forward rate. Keep today's Treasury curve and the fitted option parameters unchanged. The output is the ex-coupon value at year 2 of the remaining payments, not today's price or a holding-period return.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

14 Mini Project

Project brief. Use the fitted Hull–White model and the year-2 bond in this demo. Produce a memo with (1) the largest initial-curve repricing error, (2) the fitted mean reversion and volatility, (3) year-2 ex-coupon values for short-rate states 1 percentage point below and above the central state, and (4) the value of $5 million face in each state. Explain why these are scenario values rather than expected physical returns.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

Short rate (%)	Price per 100	Value of $5mm face
3.2817	99.8666	4,993,328.512
4.2817	96.106	4,805,300.6097
5.2817	92.4944	4,624,719.0197

```
Mean reversion a              1.7899e-01
Volatility sigma              1.1919e-02
Largest initial-curve error   1.1102e-16
dtype: float64
```

The initial curve pins down today's discounting; option premiums identify the model's risk-neutral dynamics. A higher year-2 short rate lowers the price of the same remaining payments. The $5 million position scales the per-100 result by 50,000. These states are not physical forecasts or investment returns: the two earlier coupons and their reinvestment are absent from this ex-coupon valuation. Calibration here uses synthetic option quotes, so the fitted parameters demonstrate recovery of synthetic inputs, not a live market fit.